



Nistula Summer 2026 – Your 48-Hour Technical Assessment

1 message

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To: rajeshmishra847410@gmail.com

Tue, 12 May, 2026 at 6:49 pm

Dear Rajesh Kumar Mishra,

Your application to the Nistula Summer Technology Internship 2026 has been shortlisted. This email contains your complete technical assessment. Your 48-hour window starts now.

Read this brief fully before writing a single line of code.

NISTULA TECHNICAL ASSESSMENT – 48-HOUR BRIEF

WHAT WE ARE ASSESSING

- We are not looking for perfect code. We are looking for three things:
- Can you take a real problem, break it into parts, and build something that works?
 - How do you think when something breaks or is unclear?
 - Is your code something another developer could read, understand, and build on?

We will read your code. We will read your README. We will read how you explain your decisions. All three matter equally.

PART 1 – BUILD A GUEST MESSAGE HANDLER (6-8 hours)

THE SCENARIO

Nistula receives guest messages from multiple channels. Build a backend system that:

1. Receives an inbound guest message via a webhook endpoint
2. Normalises it into a unified schema regardless of source channel
3. Sends it to the Claude API with relevant context and gets a drafted reply
4. Returns the drafted reply with a confidence score

THE WEBHOOK ENDPOINT

Build a POST endpoint at /webhook/message that accepts this payload:

```
{
  "source": "whatsapp",
  "guest_name": "Rahul Sharma",
  "message": "Is the villa available from April 20 to 24? What is the rate for 2 adults?",
  "timestamp": "2026-05-05T10:30:00Z",
  "booking_ref": "NIS-2024-0891",
  "property_id": "villa-b1"
}
```

Source can be: whatsapp, booking_com, airbnb, instagram, direct

THE UNIFIED SCHEMA

Normalise every message into this before passing to AI:

```
{
  "message_id": "generated UUID",
  "source": "whatsapp",
  "guest_name": "Rahul Sharma",
  "message": "Is the villa available from April 20 to 24? What is the rate for 2 adults?",
  "timestamp": "2026-05-05T10:30:00Z",
  "booking_ref": "NIS-2024-0891",
  "property_id": "villa-b1"
}
```

```
"message_text": "Is the villa available...",
"timestamp": "2026-05-05T10:30:00Z",
"booking_ref": "NIS-2024-0891",
"property_id": "villa-b1",
"query_type": "pre_sales_availability"
}
```

QUERY CLASSIFICATION

Classify every message into one of these types:

- pre_sales_availability: Is the villa available on these dates?
- pre_sales_pricing: What is the rate for 2 adults 3 nights?
- post_sales_checkin: What time can we check in? WiFi password?
- special_request: Early check-in, airport transfer?
- complaint: The AC is not working. I am not happy.
- general_enquiry: Do you allow pets? Is there parking?

MOCK PROPERTY CONTEXT — use this in your Claude prompt:

Property: Villa B1, Assagao, North Goa
Bedrooms: 3 | Max guests: 6 | Private pool: Yes
Check-in: 2pm | Check-out: 11am
Base rate: INR 18,000 per night (up to 4 guests)
Extra guest: INR 2,000 per night per person
WiFi password: Nistula@2024
Caretaker: Available 8am to 10pm
Chef on call: Yes, pre-booking required
Availability April 20-24: Available
Cancellation: Free up to 7 days before check-in

CLAUDE API ACCESS

We have provided a temporary API key for this assessment only:

sk-ant-api03-bgkw66BJ5AWWygndnd_9Uw3LshbqhBxemByLS8WNvbUTCXEEhFP9xAawlwFLcn_zzyqV7CnbmqHiPo6-IgJZ3BQ-K47v9AAA

This key is valid for 72 hours from this email only. Do not share it or use it for anything other than this assessment. It will be revoked after submissions close.

Use model: claude-sonnet-4-20250514

CONFIDENCE SCORING

Return a confidence score between 0 and 1 alongside the reply. Define your own logic and explain it in the README.

YOUR ENDPOINT SHOULD RETURN:

```
{
  "message_id": "uuid",
  "query_type": "pre_sales_availability",
  "drafted_reply": "Hi Rahul! Great news...",
  "confidence_score": 0.91,
  "action": "auto_send"
}
```

Action field: auto_send (above 0.85), agent_review (0.60-0.85), escalate (below 0.60 or complaint)

CONSTRAINTS

- Use Python (FastAPI) or Node.js (Express)
- API key goes in .env — never hardcode it
- Handle errors gracefully
- Test with at least 3 different inputs before submitting

PART 2 — DATABASE SCHEMA (2-3 hours)

Design the PostgreSQL schema for the Nistula unified messaging platform. Write actual SQL CREATE TABLE statements. The schema must support:

1. Guest profiles — one record per guest across all channels
2. All messages across all channels in one table
3. Conversations linked to guests linked to reservations
4. Tracking whether a message was AI drafted, agent edited, or auto-sent
5. AI confidence score and query type stored per inbound message

Submit: SQL CREATE TABLE statements + comments on design decisions + a short paragraph on the hardest design decision you made and why.

PART 3 — THINKING QUESTION (1-2 hours)

SCENARIO: It is 3am. A guest at Villa B1 sends a WhatsApp message: "There is no hot water and we have guests arriving for breakfast in 4 hours. This is unacceptable. I want a refund for tonight."

Answer these three questions. Maximum 400 words total.

Question A — The Immediate Response

What should the AI reply right now at 3am? Write the actual message. Explain in 2-3 lines why you chose this wording.

Question B — The System Design

What should the platform do beyond sending a message? Walk through the full system response: what gets triggered, who gets notified, what gets logged, what happens if no human responds within 30 minutes.

Question C — The Learning

This is the third time in two months a guest has complained about hot water at Villa B1. What should the system do with this pattern? What would you build to prevent this complaint from happening a fourth time?

SUBMISSION INSTRUCTIONS

1. Create a public GitHub repository: nistula-technical-assessment
2. Repository must contain:
 - README.md: setup instructions and confidence scoring logic explained
 - /src or /app: your Part 1 webhook code
 - schema.sql: Part 2 SQL with comments
 - thinking.md: Part 3 written answers
 - .env.example: required variables template — no actual keys
3. Reply to this email with your GitHub repository link
4. Deadline: 48 hours from the time this email was sent

DO NOT include your actual API key anywhere in the repository.

If you get stuck: write a comment explaining what you tried, what broke, and how you would approach it with more time. We read this.

Questions? Reply to this email. We respond within a few hours.

Good luck. We are looking forward to seeing what you build.

Warm regards,
Ash Verma
Founder, Nistula
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